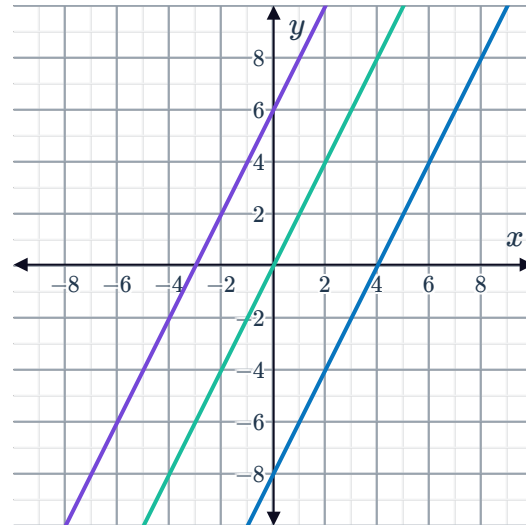


Equations of parallel and perpendicular lines



- 1 The equations $y = 2x$, $y = 2x + 6$ and $y = 2x - 8$ have been graphed on the same number plane:

- a Describe the three equations in terms of their slope.
- b Determine whether the following statements describe the given graphs.
 - i All graphs cut the y -axis at the same point.
 - ii All graphs cut the x -axis at the same point.
 - iii All graphs have the same angle of inclination.
- c What can you conclude from the answers above?



- 2 Determine whether the following pairs of lines are parallel.

- a $y = 4x - 1$ and $y = 4x - 6$
- b $y = -8x - 2$ and $y = 9x + 7$
- c $y = -6x - 5$ and $y = x$

- 3 Determine whether the following lines are parallel to $y = 9x + 2$.

- a $y = 9x$
- b $y = 5x + 5$
- c $y = 9x + 5$
- d $y = -9x + 2$
- e $y = 9x - 2$
- f $y = 5x + 2$

- 4 Given that the line formed by Equation 1 is parallel to the line formed by Equation 2, find the missing value for each of the following.

- a Equation 1: $y = 1x + 4$
Equation 2: $y = -3x - 7$
- b Equation 1: $y = 1x + 4$
Equation 2: $y = \frac{4}{5}x - 7$

- 5 Find the two equations of the line that are both parallel to $y = -4$, and 2 units away from the line $y = -4$.

Determine the equation of the original line.



- 7 Find the equation of the following lines, expressing your answer in slope-intercept form when appropriate.
- a A line that is parallel to the x -axis and passes through $(-8, 5)$
 - b A line that is parallel to the y -axis and passes through $(-7, 5)$
 - c A line that is parallel to $y = -\frac{7x}{6} - 4$ and passes through the point $(0, -2)$
 - d A line that is parallel to $y = -8x - 6$ and crosses the y -axis at $(0, 7)$
 - e A line that is parallel to $y = 2x - 1$ and passes through the point $(1, 3)$
 - f A line that is parallel to $y = \frac{2x}{3} - 10$ and passes through $(-3, -3)$
- 8 Find the equation of the line that is parallel to $y = \frac{4x}{3} - 3$ and passes through $(-2, -5)$. Express your answer in standard form.
- 9 A line passes through points $A(4, 3)$ and $B(8, -10)$.
- a Find the slope of the line.
 - b Find the equation of a line that passes through $(-1, 2)$ and is parallel to the line passing through $A(4, 3)$ and $B(8, -10)$. Express the equation in slope-intercept form.
- 10 A line passes through $A(2, -4)$ and $B(3, 9)$.
- a Find the slope of this line.
 - b Find the equation of a line that has a y -intercept of 5 and is parallel to the line that goes through $A(2, -4)$ and $B(3, 9)$. Express the equation in slope-intercept form.
- 11 The lines $y = -5mx + 1$ and $y = -2 + 4x$ are parallel. Find the value of m .

12 For each of the following:



- i Find the slope of the line.
 - ii Find the value of b if the equation is to be written in the form $y = mx + b$.
 - iii Write the equation of this line in slope-intercept or standard form.
-
- a A line passes through the point $(9, 4)$ and is parallel to the line with equation $y = 4x + 2$
 - b A line passes through the point $(8, -8)$ and is parallel to the line with equation $y = -x + 5$
 - c A line passes through the point $(-6, -4)$ and is parallel to the line with equation $y = 5x - 4$

13 Find the negative reciprocal of the following:

- | | |
|-----------------|-------------------|
| a $\frac{5}{2}$ | b $-\frac{10}{9}$ |
| c 3 | d $-\frac{1}{4}$ |

14 Determine which is the single best definition that describes two perpendicular lines.

- Two lines that have no intersections.
- Two lines that have only one intersection.
- Two lines whose product of slopes is equal to 1.
- Two lines that intersect each other at an angle of 90° .

15 Given that two non-vertical lines are perpendicular, state the product of their slopes.

16 State whether the following pairs of lines are perpendicular.

- | | |
|--|---|
| a $y = x + 6$ and $y = x - 6$ | b $y = 6x - 1$ and $y = -6x - 4$ |
| c $y = 4x - 8$ and $y = -\frac{x}{4} + 3$ | d $y = \frac{5x}{4} + 5$ and $y = 9 - \frac{4x}{5}$ |
| e $y = \frac{x}{3} - 7$ and $y = -(3x + 21)$ | f $y = 7x - 5$ and $y = -7x + 6$ |
| g $y = -8x - 3$ and $y = \frac{x}{8} - 2$ | |

17 Find the slope of the following lines.



- a A line perpendicular to a line which has a slope of -9
- b A line perpendicular to $y = -\frac{x}{3} - 7$
- c Any line perpendicular to $y = -5x + 6$
- d Any line that is perpendicular to $y = 10 - x$

18 Given that the line formed by Equation 1 is perpendicular to the line formed by Equation 2, find the missing value for each of the following.

a Equation 1: $y = 10x - 6$
Equation 2: $y = \text{lx} + 7$

b Equation 1: $y = \frac{3}{4}x + 2$
Equation 2: $y = \text{lx} + 1$

c Equation 1: $y = -10x + 4$
Equation 2: $y = \text{lx} - 6$

d Equation 1: $y = \frac{2}{5}x - 10$
Equation 2: $y = \text{lx} - 5$

19 Determine whether the following lines are perpendicular to $y = 4x - 3$.

a $y - \frac{x}{4} = \frac{3}{2}$

b $4x + 16y = 96$

c $x + 4y = -3$

d $y = -4x - 3$

e $y = \frac{x}{4} - 3$

20 Determine whether the following is an equation of the line that is perpendicular to $3y = 8 + \frac{3x}{4}$ and passes through $(-4, 8)$.

a $y = \frac{x}{4} + 9$

b $y = \frac{x}{4} - 6$

c $y = -4x + 28$

d $y = -4x - 8$

21 Write down the equation of the following lines, expressing your answers in standard form.

a A line that is perpendicular to the x -axis and passes through $(10, 3)$

b A line that is perpendicular to the y -axis and passes through $(1, -7)$

c A line that is perpendicular to $y = 7x - 7$ and passes through $(-1, 1)$

d A line that is perpendicular to $y = \frac{7x}{4} + 9$ and passes through $(4, -4)$

e A line that is perpendicular to $2x - 7y = -5$, and passes through $(-5, 2)$

22 A line L is perpendicular to the segment joining  $B(2, -10)$ and $C(-1, 10)$.

- a** Find the slope of line L .
- b** Find the equation of line L if it passes through $(-8, -4)$. Express the equation in slope-intercept or standard form.

23 Consider the line $4x - 3y = -6$.

- a** Find the y -intercept of the line.
- b** Find the equation of the line that is perpendicular to the given line and has the same y -intercept. Express the equation in standard form.

24 Consider the line that is perpendicular to $3x - 8y = -2$ and passes through $(1, 2)$.

- a** Find the equation of the line in standard form.
- b** Does the point $\left(-3, \frac{38}{3}\right)$ lie on the line?

25 Consider the line that is perpendicular to $5x + 2y = -7$ and passes through $(4, 0)$.

- a** Find the equation of the line in standard form.
- b** Does the point $\left(-4, -\frac{11}{5}\right)$ lie on the line?

26 The line L_1 is perpendicular to $y = 5x + 6$ and intersects the y -axis at 6.

- a** Determine the slope of L_1 .
- b** State the equation of L_1 , writing your answer in slope-intercept form.

27 Find the equation of a line that is perpendicular to $y = 10x + 3$, and has the same y -intercept. Express the equations in slope-intercept or standard form.

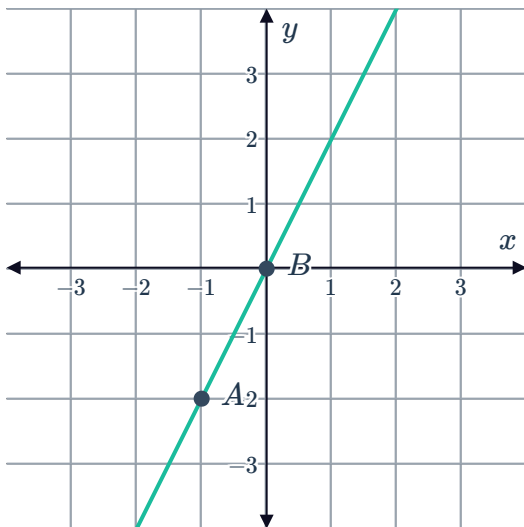


Additional questions

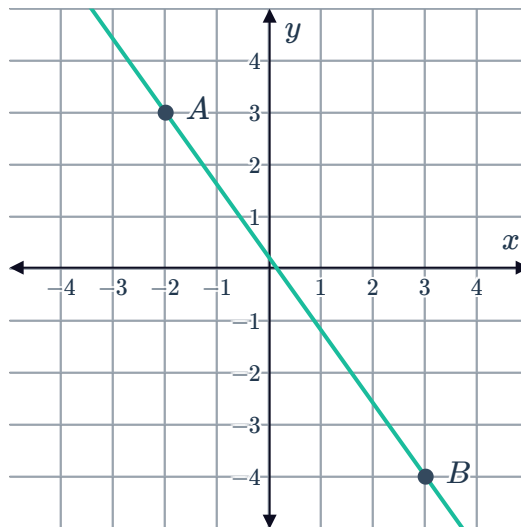
28 For each of the following:

- i Write the slope of a line that would be parallel to the graphed line.
- ii Write the slope of a line that would be perpendicular to the graphed line.

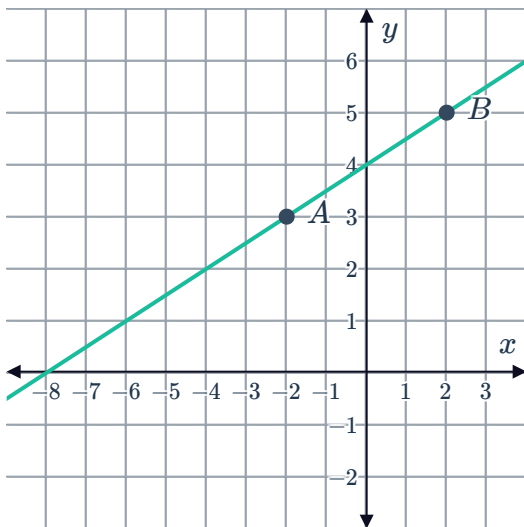
a



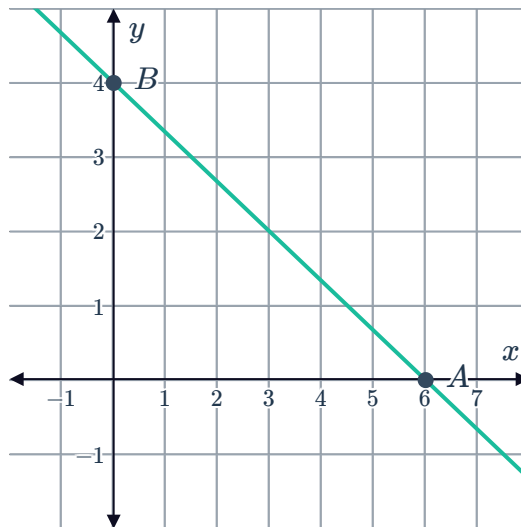
b



c



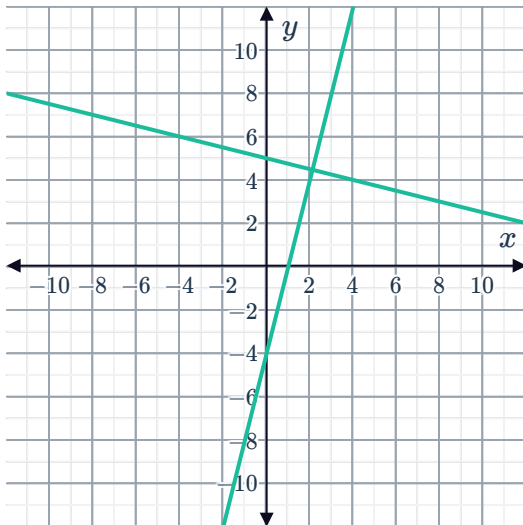
d



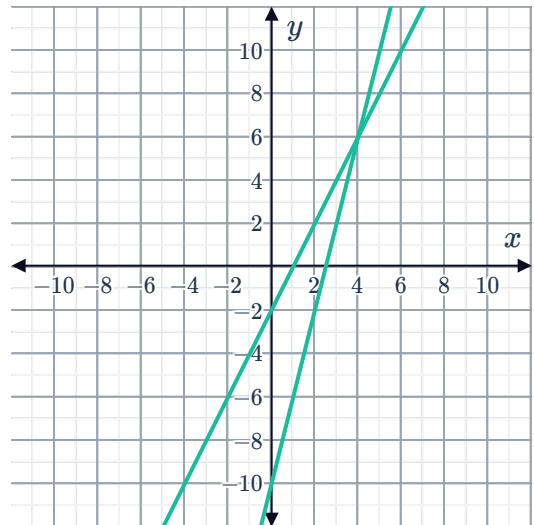
29 Determine whether the following lines are parallel, perpendicular, or neither.



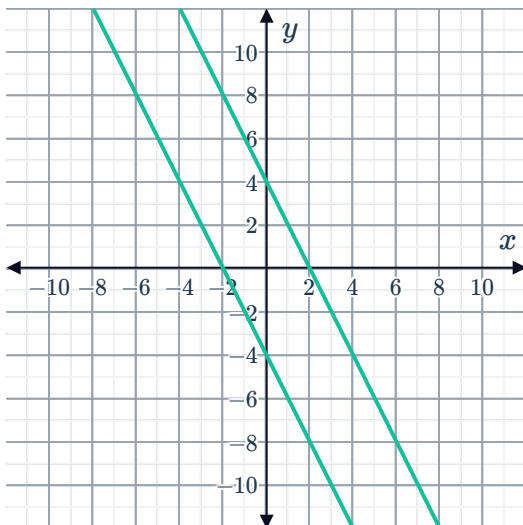
a



b



c



30 State the relationship between the slopes of

- a Parallel lines.
- b Perpendicular lines.

31 Find the slope, m , of a line that is

- a Parallel to a line which has a slope of 4.
- b Parallel to $y = -\frac{x}{3} - 7$.
- c Perpendicular to a line which has a slope of -9 .
- d Perpendicular to $y = -5x + 6$.



32 Determine if the following lines are parallel to $y = \frac{2}{3}x - 3$ or not.

a $y = \frac{2}{3}x + 1$

b $4x - 6y = 37$

c $3x + 2y = -13$

d $y = -\frac{2}{3}x - 8$

33 Determine if the following lines are perpendicular to $y = 4x - 3$ or not.

a $y = \frac{x}{4} - \frac{3}{2}$

b $4x + 16y = 96$

c $x + 4y = -3$

d $y = -4x - 3$

34 For each of the following pair of lines, determine whether they are parallel, perpendicular, or neither.

a $y = 3x - 7$ and $y = 3x + 4$

b $x - y = 6$ and $x - y = -6$.

c $8x + y = 2$ and $y = \frac{x}{8} - 5$

d $2x + 3y = 5$ and $-3x - 2y = 12$

35 Find the equation the line that is

a Parallel to the line $y = -\frac{7x}{6} - 4$ and passes through the point $(0, -2)$.

b Parallel to $x = 1$ and passes through $(-4, 6)$.

c Parallel to the x -axis and passes through $(-8, 5)$.

d Parallel to the y -axis and passes through $(-7, 5)$.

36 Find the equation the line that is

a Perpendicular to $x = 4$ and passes through $(-7, 1)$.

b Perpendicular to $y = 2x - 1$ and passes through the point $(2, 3)$.

c Perpendicular to the x -axis and passes through the point $(10, 3)$.

d Perpendicular to the y -axis and passes through $(1, -7)$.

37 Consider a line that passes through points $A(4, 3)$ and $B(8, -10)$.

a Find the slope, m , of the line.

b Find the equation of a line that passes through $(-1, 2)$ and is parallel to the line passing through $A(4, 3)$ and $B(8, -10)$. Express the equation in slope-intercept form.



38 Consider a line that passes through $A(2, -4)$ and $B(3, 9)$.

- a** Find the slope, m , of the line.
- b** Find the equation of a line that has a y -intercept of 5 and is perpendicular to the line that goes through $A(2, -4)$ and $B(3, 9)$. Express the equation in slope-intercept form.

39 Consider line L_1 with equation $3x - 4y - 4 = 0$.

- a** Write L_1 in slope-intercept form.
- b** Find the slope of a line, L_2 , that is perpendicular to L_1 .
- c** Find the equation of L_2 in standard form, given that it passes through the point $A(-6, 4)$.

40 Consider the lines $2x - 7y = 2$ and $mx - 35y = -7$.

Given that the two lines are parallel, determine the value of m .

41 Explain why two lines with the standard forms $ax + by = c_1$ and $bx - ay = c_2$ are perpendicular.

42 Ammon and Minh work part-time at the same pizzeria, both earning \$360 per week. Both Ammon and Minh are saving up, so they don't spend any of the money they earn. Ammon currently has a bank balance of \$212 while Minh has \$275.

If we sketch linear graphs for Ammon and Minh's bank balances over time, can we expect to get two parallel lines? Justify your answer.

43 Suppose that the lines $px - 4y = 7$ and $qx + 5y = -14$ are perpendicular.

Write an equation relating the values of p and q .